

MACHINE LEARNING METHODOLOGIES

SUMMARY III (Version 2)

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This summary contains what I understand from the second resource on a Machine Learning Methodology; **FEATURE SELECTION**.

Need for Feature Selection

When a machine learning model is trained with all available features, the model may not always achieve good performance. Some features may be irrelevant, redundant, or only weakly related to the target.

Feature selection helps address this by identifying a subset of the available features that are more useful for the model. By removing unnecessary features and retaining the more informative ones, the model may achieve better predictive performance and accuracy.

However, feature selection does not automatically guarantee higher accuracy; its effectiveness depends on the dataset, features, target, and model being used.

What is Feature Selection?

Feature Selection is the process of identifying and selecting the most relevant features for a machine learning model while removing irrelevant or less useful features from the dataset.

The objective is to provide the model with useful information without unnecessarily increasing the number of variables it has to process.

Importance of Feature Selection

Feature selection is important for several reasons:

Reduces the Curse of Dimensionality

As the number of features increases, the feature space becomes increasingly complex. This is known as the **Curse of Dimensionality**.

Selecting only relevant features helps reduce the dimensionality of the dataset and can make the learning problem more manageable.

Reduces Computational Cost

Fewer features mean less data for the model to process. This can reduce computational requirements and potentially shorten training time.

Supports Model Learning

Removing irrelevant features can help the model focus on the variables that contain useful information for predicting the target.

Can Improve Model Accuracy

When irrelevant or noisy features are removed, the model may be able to learn the underlying patterns more effectively, which can lead to improved predictive performance.

Feature Selection Statistics

The appropriate statistical technique for feature selection depends on the type of variables involved.

Both the **input variables (features)** and the **output variable (target)** can be either:

- Categorical
- Numerical

Therefore, the nature of the input and output variables helps determine which statistical technique is appropriate for evaluating their relationship.

Feature Selection Techniques

Feature selection can broadly be divided into:

Supervised Feature Selection

Supervised feature selection uses the **target or output variable** when determining which features are relevant.

The relationship between the input features and the target is considered during selection.

Common supervised approaches include:

- Filter Methods
- Wrapper Methods
- Embedded Methods

Unsupervised Feature Selection

Unsupervised feature selection does **not require the target variable**.

The selection process is based on characteristics of the input features themselves rather than their relationship with a known output.

Filter Methods

Filter methods select features using statistical characteristics or relationships within the data, usually before a machine learning model is trained.

Some techniques include:

Information Gain

Information Gain measures how much uncertainty, or **entropy**, is reduced when information from a feature is used.

A feature that produces a greater reduction in entropy provides more information about the target.

Chi-Squared Test

The Chi-squared test is used to examine the relationship or association between **categorical variables**.

It can therefore be useful when evaluating categorical features against a categorical target.

Fisher's Score

Fisher's Score evaluates the discriminative ability of features and can be used to rank features according to their usefulness for distinguishing between classes.

Features can then be ranked in descending order of their scores.

Missing Value Ratio

The Missing Value Ratio examines the proportion of missing values in each feature.

Features with a missing-value ratio above a chosen threshold can be considered for removal.

Wrapper Methods

Wrapper methods select features by repeatedly evaluating different feature subsets using a **machine learning model or classifier**.

Unlike filter methods, wrapper methods directly involve the model when determining which features are useful.

Forward Selection

Forward Selection begins with an **empty set of features**.

Features are gradually added one at a time. After each addition, the model is evaluated to determine whether the new feature improves performance.

The process continues until an appropriate feature subset is obtained.

Backward Selection

Backward Selection starts with **all available features**.

Features are progressively removed based on their contribution to the model.

The least significant or least useful feature is removed at each stage, followed by another model evaluation.

Exhaustive Feature Selection

Exhaustive Feature Selection evaluates different possible combinations of features to identify the subset that provides the best result according to the chosen evaluation criterion.

Because it considers many possible combinations, it can become computationally expensive as the number of features increases.

Recursive Feature Elimination (RFE)

RFE is a **recursive, greedy feature-selection approach**.

It begins with the available features, trains a model, evaluates feature importance, and removes the least important features. The process is repeated until the desired number of features remains.

Embedded Methods

Embedded methods perform feature selection as part of the model-training process.

They therefore combine aspects of the ideas behind filter and wrapper approaches: feature selection is connected to the learning algorithm, but it is incorporated directly into the model-training process.

Common examples include:

Regularization

Regularization methods can encourage models to reduce the influence of unnecessary features.

Examples include:

- **L1 Regularization (Lasso)**
- **L2 Regularization (Ridge)**
- **Elastic Net**

L1 regularization is particularly associated with feature selection because it can drive some feature coefficients to zero.

Tree-Based Methods

Decision trees and other tree-based models can provide measures of feature importance.

Examples include:

- Decision Trees
- Random Forests
- Other tree-based methods

These important measures can be used to identify features that contribute more strongly to the model.

Advantages and Disadvantages of Feature Selection Methods

Filter Methods

Advantages

Better computational complexity

They are generally less computationally demanding because they can evaluate features without repeatedly training a model.

Independent of the classifier

Feature selection is not tied to a particular machine learning algorithm.

Fast and scalable

They are generally suitable for datasets with a large number of features.

Disadvantages

Potential vagueness

Statistical relevance does not always translate directly into usefulness for a particular machine learning model.

May ignore feature dependencies

Individual features may be evaluated separately even when combinations of features are important.

Does not consider classifier interaction

The method does not directly account for how a particular model will use the selected features.

Wrapper Methods

Advantages

Interact with the classifier

The feature-selection process directly considers model performance.

Model-dependent selection

Features are selected based on how useful they are to the particular model being evaluated.

Can identify useful feature combinations

Because subsets are evaluated together, wrapper methods can capture interactions between features more effectively than methods that evaluate each feature independently.

Disadvantages

Computationally intensive

Repeatedly training and evaluating models can require significant computational resources.

Higher risk of overfitting

Because the selection process is closely connected to model performance, there is a greater risk of selecting features that perform well on the data used during selection but do not generalize well.

Embedded Methods

Embedded methods provide a middle ground between filter and wrapper approaches.

Feature selection occurs **during the model-training process**, rather than as a completely separate preprocessing step.

This allows embedded methods to benefit from the model's learning process while generally being more computationally efficient than methods that repeatedly evaluate many separate feature subsets.

However, embedded methods are typically **model-dependent**, meaning their selected features can depend on the algorithm being used.

Overall Comparison

The three approaches can be remembered by asking **when and how the features are selected**:

Filter Method

Features are selected using statistical properties or relationships, independently of a specific machine learning model.

Wrapper Method

Different feature subsets are evaluated through a machine learning model to determine which subset performs best.

Embedded Method

Feature selection happens as part of the model-training process.

Key Takeaway

Feature selection is about finding the **most useful set of features** for a machine learning task rather than simply using every available column.

It can help reduce dimensionality, lower computational requirements, simplify the learning process, and potentially improve model performance.

The three major approaches are:

Filter → Statistical evaluation

Wrapper → Model-based evaluation

Embedded → Selection built into model training

The choice of method depends on factors such as the type of data, number of features, computational resources, machine learning algorithm, and the objective of the analysis.